

Geometry
Unit 7
Lesson H3 Practice

Name Answer Key
Date _____
Period _____

Determine the area of each triangle using the given information. Round to the nearest tenth.

	Triangle	Given Information			Area
1.	$\triangle BHZ$	$HB = 7$	$ZB = 12$	$m\angle HBZ = 64^\circ$	$= \frac{1}{2}(7)(12)(\sin 64)$ $= 37.7 \text{ units}^2$
2.	$\triangle YCJ$	$CY = 100$	$JY = 63$	$m\angle JYC = 80^\circ$	$= \frac{100 \cdot 63 \cdot \sin 80}{2}$ $= 3,102.1 \text{ units}^2$
3.	$\triangle KDN$	$DN = 7.2$	$NK = 6.3$	$m\angle DNK = 95^\circ$	$= \frac{1}{2}(7.2)(6.3) \sin 95$ $= 22.6 \text{ units}^2$
4.	$\triangle FSG$	$SF = 3\sqrt{5}$	$GS = 2\sqrt{7}$	$m\angle FSG = 36^\circ$	$= \frac{1}{2}(3\sqrt{5})(2\sqrt{7})(\sin 36)$ $= 10.4 \text{ units}^2$

5.	$\triangle WQL$	$QW = 8.4$	$LW = 9.9$	$m\angle QWL = 68^\circ$	$= \frac{1}{2}(8.4)(9.9)(\sin 68)$ $= 38.6 \text{ units}^2$
6.	$\triangle BRS$	$SR = \frac{78}{5}$	$RB = \frac{119}{8}$	$m\angle SRB = 42^\circ$	$= \frac{1}{2}\left(\frac{78}{5}\right)\left(\frac{119}{8}\right)(\sin 42)$ $= 77.6 \text{ units}^2$
7.	$\triangle TXD$	$TX = 2.6$	$TD = 5.1$	$m\angle XTD = 16^\circ$	$= \frac{1}{2}(2.6)(5.1) \sin 16$ $= 1.8 \text{ units}^2$
8.	$\triangle YNK$	$NY = \sqrt{11}$	$NK = \sqrt{11}$	$m\angle KNY = 72^\circ$	$= \frac{1}{2}(\sqrt{11})(\sqrt{11}) \sin 72$ $= 5.2 \text{ units}^2$
9.	$\triangle LGR$	$GL = 521$	$LR = 94$	$m\angle RLG = 179^\circ$	$= \frac{1}{2}(521)(94) \sin 179$ $= 427.4 \text{ units}^2$
10.	$\triangle ABC$	$BC = 5\sqrt{3}$	$AB = 4\sqrt{7}$	$m\angle ABC = 89^\circ$	$= \frac{1}{2}(5\sqrt{3})(4\sqrt{7}) \sin 89$ $= 45.8 \text{ units}^2$